MANIPAL ACADEMY OF HIGHER EDUCATION
Institution of Embrace Diamant in the University

HackSprint

24-Hour Hackathon

17-18 OCTOBER

VENUE DETAILS
Manipal Academy of Higher Education (MAHE)

SATURDAY
09 A.M. Onwards

YOUR NAME: Lalithraj B H

TEAM NAME: NextGenCoders

TRACK NO:Ps21



REGISTER NOW

Team Size
1-4 Members

ORGANIZED BY:
commudle
MANIPAL ACADEMY OF HIGHER EDUCATION

Solution

A transparent pricing loop that replaces intuition with deterministic financial logic and explainable AI.

01.

Pricing Engine

Deterministic rules for baseline price, financial bands, and margin guardrails. No AI guesswork in numbers.

02.

Market Context

Real-time comparison with competitor prices to ensure the merchant stays competitive while protecting margins.

03.

Explainable QR

Controlled AI rephrases recorded cost changes into clear explanations for customers to build long-term trust.

Tech Stack & Architecture

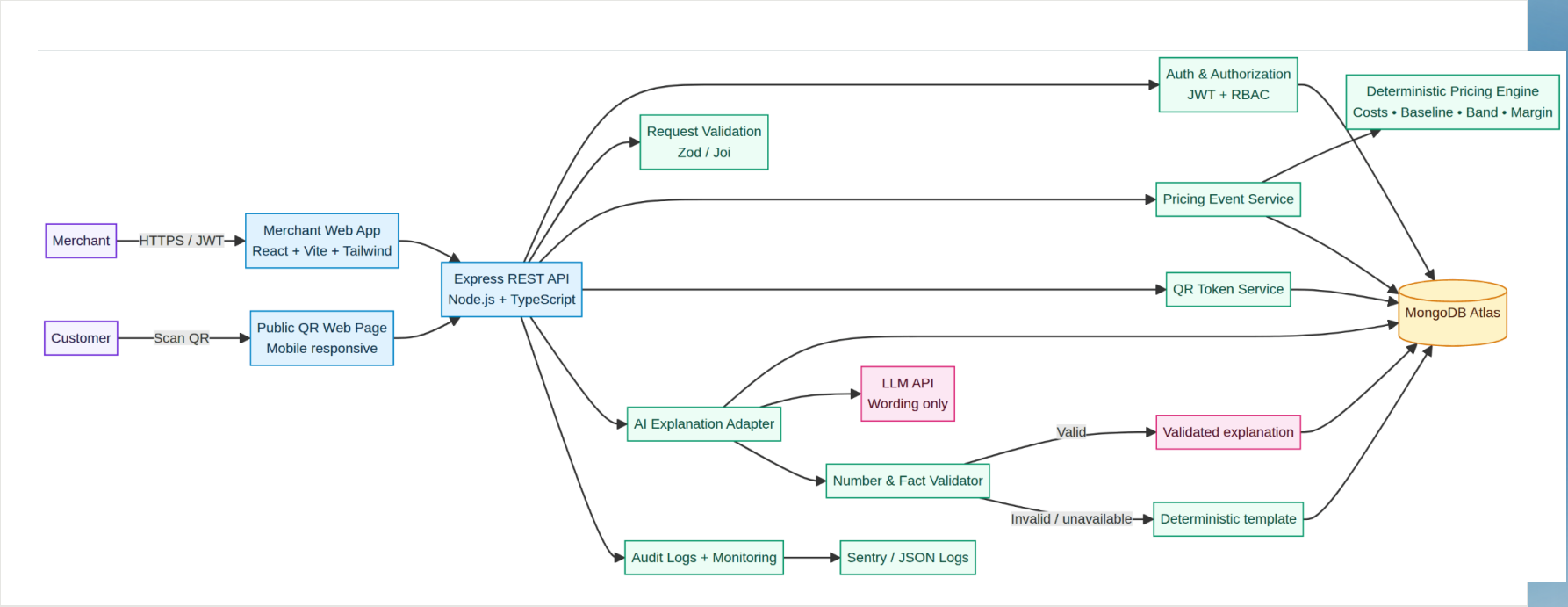
UI/UX	**React + Vite + Tailwind**
API	**Node.js + Express + TypeScript**
DATABASE	**MongoDB Atlas**
INTELLIGENCE	**LLM API + Validation Adapter**

Engineering Focus

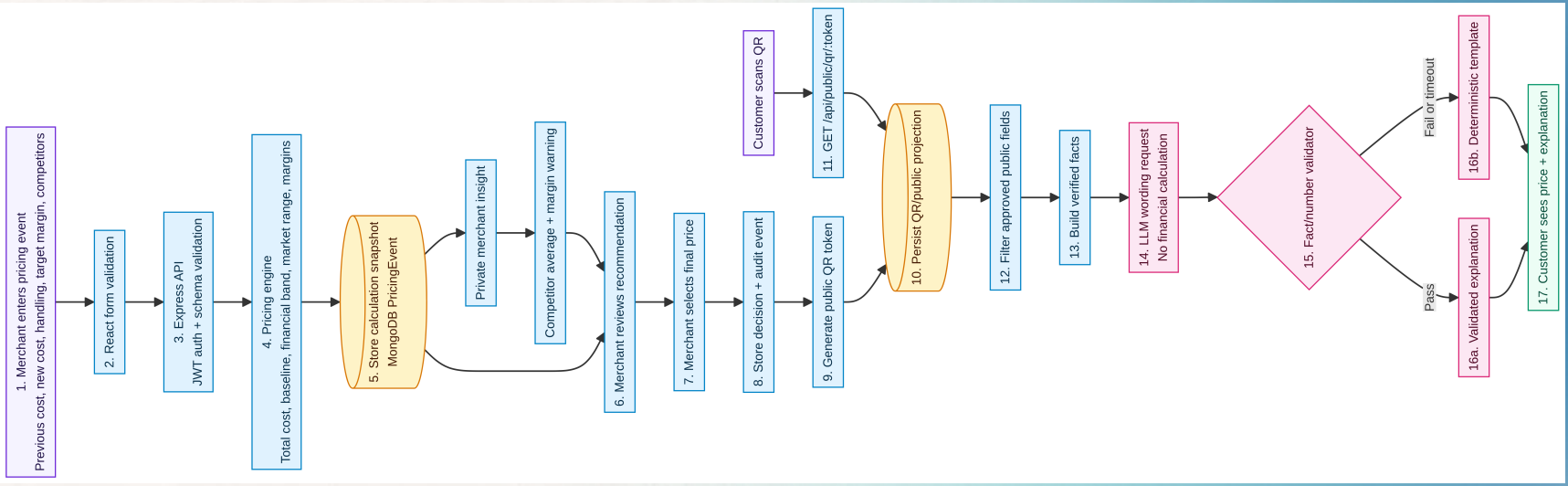
Architecture follows the **Separation of Concerns** principle: financial calculations are isolated in a pure-function engine, while AI is relegated to a controlled rephrasing adapter with strict fact-validation.

Stateless JWT authentication and MongoDB persistence allow for rapid scaling and easy merchant multi-tenancy.

Architecture

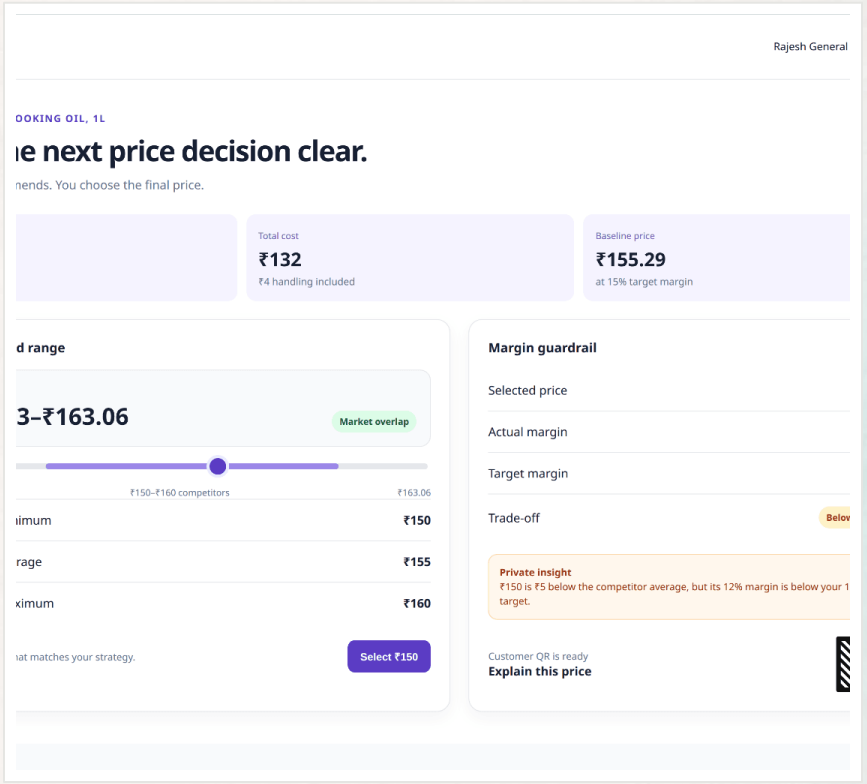


Data Flow Diagram

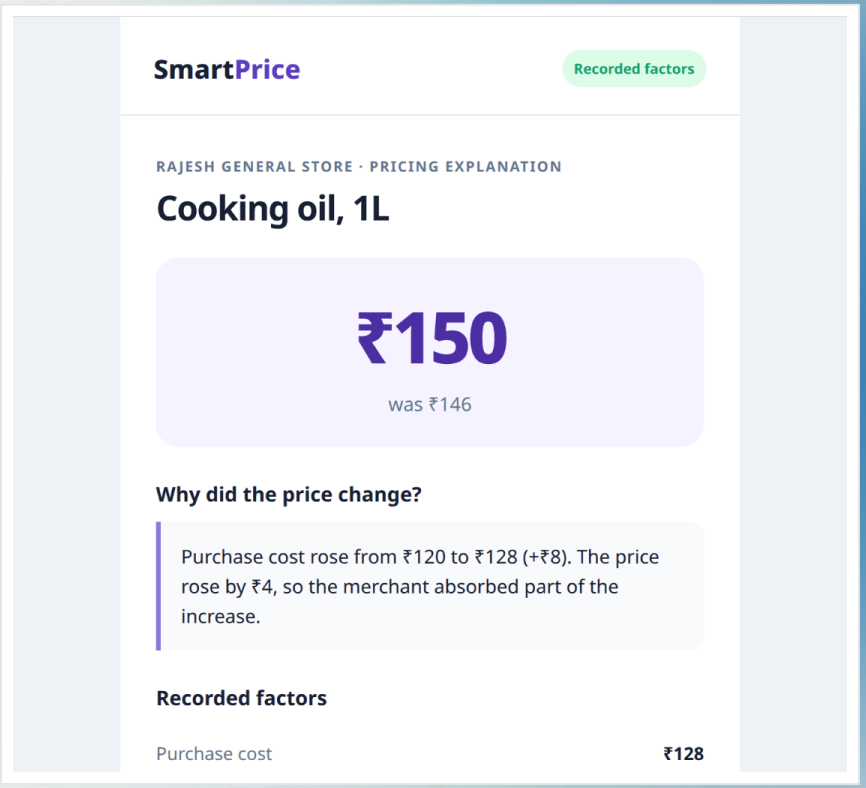


Screenshot of Your Project

MERCHANT DASHBOARD



CUSTOMER QR EXPLANATION



Others

Future Scope

Live competitor scraping for automated market context.

Demand forecasting and price sensitivity analysis.

Full WhatsApp & ERP integration for small merchants.

Automated price publishing to delivery platforms.

Business Value

Trust: Reduces customer friction during cost-driven price hikes.

Speed: Instant calculation of margins protects profit in volatile markets.

Simplicity: A lightweight layer that sits atop existing billing tools.

Scale: Merchant-agnostic platform for global small commerce.